

**MAMA, Russian Federation**

WMO: 301570

Lat: **58.313N**Lon: **112.890E**Elev: **735**StdP: **14.31**Time Zone: **8.00 (E08)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -44.8      | -41.2 | -48.6                         | 0.3 | -44.3 | -45.1 | 0.4 | -40.8 | 19.0                  | -2.9 | 16.7 | -6.4 | 2.5                   | 120  | 0.678 |

**Annual Cooling, Dehumidification, and Enthalpy Design Conditions**

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 21.7                   | 86.4            | 67.0 | 82.2 | 65.9 | 78.4 | 64.3 | 69.9                | 81.1 | 68.0 | 78.3 | 66.1 | 75.6 | 6.4                  | 150  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 66.4                            | 99.9 | 74.6 | 64.4 | 93.1 | 72.5 | 62.4 | 86.5 | 70.6 | 34.5          | 81.5 | 32.8 | 78.2 | 31.4 | 75.9 | 82.8              |

**Extreme Annual Design Conditions**

| Extreme Annual WS |             |             | Extreme Annual Temperature |              |                    |            | n-Year Return Period Values of Extreme Temperature |              |             |              |             |              |             |              |             |    |
|-------------------|-------------|-------------|----------------------------|--------------|--------------------|------------|----------------------------------------------------|--------------|-------------|--------------|-------------|--------------|-------------|--------------|-------------|----|
|                   |             |             | Mean                       |              | Standard Deviation |            | n=5 years                                          |              | n=10 years  |              | n=20 years  |              | n=50 years  |              |             |    |
| 1%                | 2.5%        | 5%          | Min                        | Max          | Min                | Max        | Min                                                | Max          | Min         | Max          | Min         | Max          | Min         | Max          | DB          | WB |
| (a)               | (b)         | (c)         | (e)                        | (f)          | (g)                | (h)        | (i)                                                | (j)          | (k)         | (l)          | (m)         | (n)          | (o)         | (p)          |             |    |
| (4)               | <b>17.2</b> | <b>14.5</b> | <b>12.8</b>                | <b>-47.9</b> | <b>93.0</b>        | <b>5.6</b> | <b>2.5</b>                                         | <b>-51.9</b> | <b>94.8</b> | <b>-55.2</b> | <b>96.3</b> | <b>-58.3</b> | <b>97.7</b> | <b>-62.4</b> | <b>99.5</b> |    |
| (5)               |             |             |                            | <b>-47.5</b> | <b>74.4</b>        | <b>5.1</b> | <b>2.6</b>                                         | <b>-51.2</b> | <b>76.3</b> | <b>-54.2</b> | <b>77.9</b> | <b>-57.1</b> | <b>79.3</b> | <b>-60.8</b> | <b>81.3</b> |    |

**Monthly Climatic Design Conditions**

|                                                                               |                                                                               | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |
|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                               |                                                                               | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg                                                                         | 26.2   | -17.3 | -6.3  | 11.2  | 31.1  | 45.1  | 60.5  | 66.3  | 60.5  | 45.2  | 27.9  | 2.8   | -15.0 |      |
|                                                                               | DBStd                                                                         | 31.20  | 15.53 | 14.73 | 13.80 | 8.53  | 7.05  | 7.39  | 5.90  | 6.35  | 7.02  | 9.39  | 16.30 | 15.68 |      |
|                                                                               | HDD50                                                                         | 9926   | 2085  | 1576  | 1202  | 568   | 187   | 8     | 0     | 6     | 178   | 686   | 1415  | 2015  |      |
|                                                                               | HDD65                                                                         | 14334  | 2550  | 1996  | 1667  | 1018  | 617   | 171   | 57    | 167   | 595   | 1151  | 1865  | 2480  |      |
|                                                                               | CDD50                                                                         | 1228   | 0     | 0     | 0     | 0     | 35    | 325   | 505   | 330   | 33    | 0     | 0     | 0     |      |
|                                                                               | CDD65                                                                         | 159    | 0     | 0     | 0     | 0     | 0     | 37    | 96    | 26    | 0     | 0     | 0     | 0     |      |
|                                                                               | CDH74                                                                         | 1910   | 0     | 0     | 0     | 0     | 27    | 629   | 1000  | 250   | 4     | 0     | 0     | 0     |      |
|                                                                               | CDH80                                                                         | 586    | 0     | 0     | 0     | 0     | 5     | 195   | 336   | 50    | 0     | 0     | 0     | 0     |      |
| Wind                                                                          | WSAvg                                                                         | 5.1    | 5.2   | 4.7   | 5.1   | 5.7   | 5.8   | 5.1   | 4.6   | 4.5   | 4.6   | 5.3   | 5.6   | 5.2   |      |
| Precipitation                                                                 | PrecAvg                                                                       | 21.9   | 1.1   | 0.9   | 0.9   | 1.1   | 1.6   | 2.5   | 2.6   | 2.8   | 2.6   | 2.4   | 2.0   | 1.3   |      |
|                                                                               | PrecMax                                                                       | 27.6   | 2.7   | 1.8   | 2.0   | 2.3   | 4.2   | 5.7   | 6.0   | 5.6   | 4.3   | 6.6   | 5.8   | 2.7   |      |
|                                                                               | PrecMin                                                                       | 17.8   | 0.5   | 0.4   | 0.2   | 0.5   | 0.8   | 1.0   | 0.9   | 0.9   | 1.0   | 0.6   | 0.8   | 0.5   |      |
|                                                                               | PrecStd                                                                       | 2.4    | 0.6   | 0.4   | 0.4   | 0.5   | 0.8   | 0.9   | 1.3   | 1.1   | 0.8   | 1.4   | 1.0   | 0.5   |      |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%                                                                          | DB     | 23.5  | 29.9  | 45.0  | 59.6  | 77.4  | 90.3  | 92.1  | 85.1  | 71.4  | 50.1  | 38.7  | 25.3  |      |
|                                                                               |                                                                               | MCWB   | 21.6  | 27.2  | 35.9  | 45.4  | 57.4  | 67.4  | 70.2  | 68.1  | 60.3  | 42.9  | 35.5  | 23.5  |      |
|                                                                               | 2%                                                                            | DB     | 16.3  | 25.8  | 37.8  | 53.6  | 70.7  | 85.8  | 87.7  | 80.4  | 64.1  | 45.4  | 34.9  | 19.5  |      |
|                                                                               |                                                                               | MCWB   | 14.9  | 23.4  | 31.6  | 41.8  | 53.5  | 65.0  | 68.5  | 65.8  | 54.5  | 40.0  | 33.2  | 18.1  |      |
|                                                                               | 5%                                                                            | DB     | 9.6   | 20.4  | 33.9  | 48.2  | 65.4  | 81.1  | 83.6  | 76.2  | 59.8  | 41.9  | 30.9  | 12.8  |      |
|                                                                               |                                                                               | MCWB   | 8.5   | 18.5  | 29.5  | 38.6  | 50.4  | 63.8  | 67.3  | 64.4  | 52.8  | 38.2  | 29.0  | 11.6  |      |
|                                                                               | 10%                                                                           | DB     | 4.4   | 14.6  | 30.6  | 43.7  | 60.3  | 77.0  | 79.8  | 72.1  | 56.1  | 39.4  | 25.1  | 7.3   |      |
|                                                                               |                                                                               | MCWB   | 3.5   | 13.1  | 26.4  | 36.1  | 47.9  | 62.0  | 66.2  | 62.3  | 50.6  | 36.3  | 23.5  | 6.3   |      |
|                                                                               | Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%   | WB    | 21.6  | 27.7  | 36.0  | 46.2  | 58.6  | 71.0  | 73.4  | 71.1  | 61.3  | 44.8  | 36.7  | 23.8 |
|                                                                               |                                                                               |        | MCDB  | 23.4  | 29.8  | 43.5  | 57.2  | 75.3  | 83.3  | 85.1  | 81.3  | 68.0  | 48.8  | 37.9  | 25.2 |
| 2%                                                                            |                                                                               | WB     | 15.1  | 23.5  | 32.6  | 42.6  | 54.7  | 67.8  | 70.8  | 68.0  | 56.8  | 41.4  | 33.3  | 18.1  |      |
|                                                                               |                                                                               | MCDB   | 16.5  | 25.8  | 36.9  | 51.9  | 68.0  | 80.4  | 82.2  | 76.5  | 62.0  | 43.7  | 34.8  | 19.4  |      |
| 5%                                                                            |                                                                               | WB     | 8.4   | 18.6  | 29.7  | 39.6  | 51.3  | 65.6  | 69.0  | 65.8  | 53.7  | 38.8  | 29.0  | 11.8  |      |
|                                                                               |                                                                               | MCDB   | 9.4   | 20.5  | 34.1  | 47.0  | 62.9  | 77.6  | 79.8  | 73.4  | 58.3  | 41.4  | 30.7  | 12.9  |      |
| 10%                                                                           |                                                                               | WB     | 3.5   | 13.3  | 26.5  | 36.8  | 48.5  | 63.3  | 67.4  | 63.6  | 51.2  | 36.4  | 23.6  | 6.3   |      |
|                                                                               |                                                                               | MCDB   | 4.3   | 14.7  | 30.4  | 42.7  | 59.1  | 74.4  | 77.3  | 70.6  | 55.2  | 39.0  | 25.0  | 7.2   |      |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB                                                                         | MDBR   | 11.8  | 16.6  | 23.0  | 19.2  | 21.4  | 24.2  | 21.7  | 18.9  | 14.8  | 10.6  | 11.1  | 10.6  |      |
|                                                                               |                                                                               | MCDBR  | 14.9  | 16.6  | 21.1  | 25.4  | 32.8  | 34.4  | 31.3  | 27.5  | 22.5  | 13.6  | 11.5  | 15.1  |      |
|                                                                               |                                                                               | MCWBR  | 13.9  | 14.2  | 15.8  | 14.6  | 16.7  | 15.8  | 13.8  | 13.5  | 12.7  | 8.9   | 9.8   | 14.3  |      |
|                                                                               | 5% WB                                                                         | MCDBR  | 14.8  | 16.1  | 20.6  | 23.6  | 29.8  | 28.8  | 26.0  | 23.4  | 19.2  | 12.3  | 10.6  | 15.0  |      |
|                                                                               |                                                                               | MCWBR  | 13.9  | 14.2  | 15.9  | 14.1  | 16.2  | 14.2  | 12.7  | 12.4  | 11.4  | 8.8   | 9.3   | 14.1  |      |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub                                                                          | 0.269  | 0.306 | 0.327 | 0.365 | 0.404 | 0.443 | 0.460 | 0.417 | 0.350 | 0.310 | 0.274 | 0.252 |       |      |
|                                                                               | taud                                                                          | 1.961  | 1.962 | 1.999 | 2.005 | 2.118 | 2.197 | 2.180 | 2.280 | 2.409 | 2.260 | 2.002 | 1.930 |       |      |
|                                                                               | Ebn at Noon                                                                   | 164    | 216   | 252   | 263   | 261   | 252   | 244   | 247   | 250   | 221   | 160   | 128   |       |      |
|                                                                               | Edh at Noon                                                                   | 19     | 30    | 39    | 46    | 44    | 42    | 42    | 35    | 27    | 23    | 17    | 14    |       |      |
| All-Sky Solar<br>Radiation                                                    | RadAvg                                                                        | 199    | 493   | 997   | 1486  | 1671  | 1740  | 1593  | 1276  | 765   | 383   | 216   | 124   |       |      |
|                                                                               | RadStd                                                                        | 16     | 39    | 74    | 83    | 115   | 165   | 172   | 117   | 81    | 54    | 30    | 11    |       |      |

**Historical Trends**

|                        |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------|--|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                        |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
| Station Only           |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| Regional (3 neighbors) |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
|                        |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |